PROJECT LIF: A LIP READING LEARNING TOOL FOR BASIC FILIPINO WORDS USING 3D CNN AND BIDIRECTIONAL LONG SHORT TERM MEMORY

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Abstract

Project LIF: A Lip Reading Learning Tool for Basic Filipino Words using 3D CNN and Bidirectional LSTM addresses the lack of Filipino-specific lip-reading technologies that promote inclusive communication and education. While most existing systems focus on English or other foreign languages, Filipino speakers—particularly individuals with speech and hearing impairments have limited access to localized visual speech recognition tools.

To address this gap, the researchers developed a web-based tutorial application that recognizes Filipino syllables and words for lip-reading practice and language learning. The system employs computer vision and deep learning techniques using a 3D CNN integrated with a BiLSTM model trained on a custom dataset of 80 syllables and 30 Filipino words. The model achieved high performance, with up to 99.02% training accuracy, 98.49% testing accuracy, and 97.35% validation accuracy for syllables, and similarly high results for word recognition. The study establishes a strong baseline for Filipino lip-reading research and recommends expanding the dataset and developing real-time, sentence-level capabilities to enhance its potential as an assistive and educational communication tool.

Keywords: Filipino Lip Reading, Visual Speech Recognition, 3D Convolutional Neural Network, Bidirectional Long Short-Term Memory, Accessibility in Educational Technology